

WASP-140b

R-2382

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_190123 · created 2026-08-20T01:29:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458506.6064 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.011
Transit depth (Rp/Rs)^2	2.04 +/- 0.31 [%]
Orbital Inclination (inc)	84.69 +/- 0.38
Airmass coefficient 1 (a1)	1.2675 +/- 0.0087
Airmass coefficient 2 (a2)	-0.142 +/- 0.033
Scatter in the residuals of the lightcurve fit is	0.69 %
Best Comparison Star	None
Optimal Aperture	5.9
Optimal Annulus	11.71
Transit Duration (day)	0.0592 +/- 0.0026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.011 vs 0.1432 (deviation 0.02σ)
Duration fit vs published	0.0592 d vs 0.0604 d (deviation 0.46σ)
Mid-time O–C	-3.4 min
Depth SNR	12.8
Residual scatter	0.69 %

Light curve

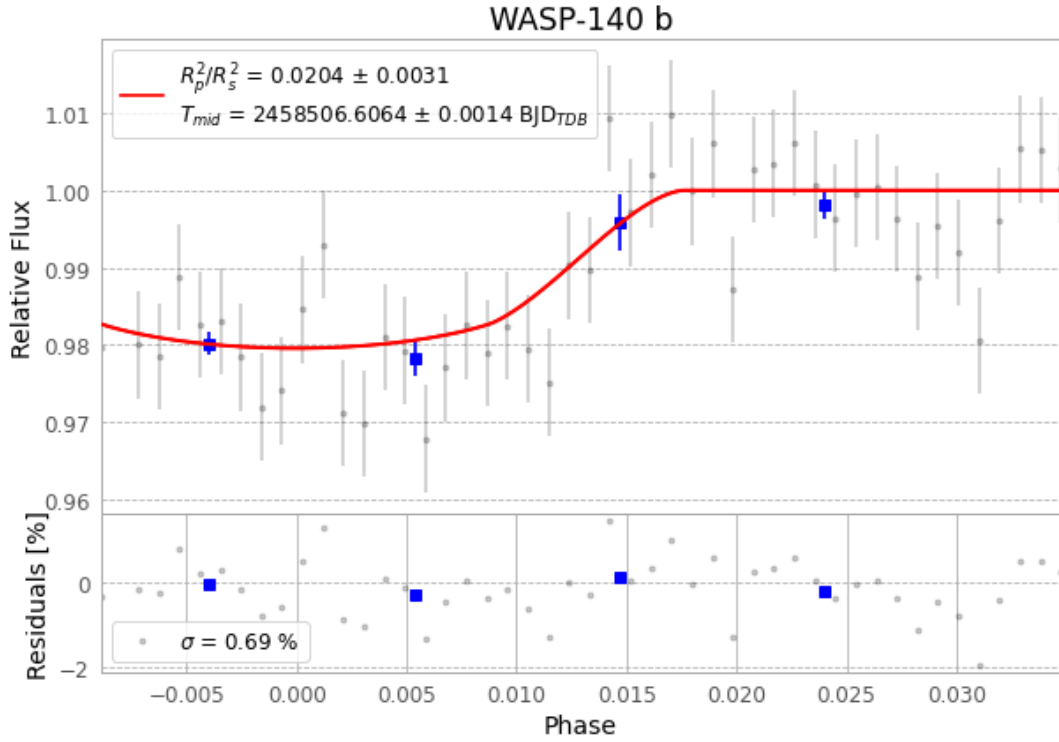


Figure 1 — FinalLightCurve_WASP-140 b_2019-01-22.png (SHA-256 7a6df098a722cb1f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [386, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9eed93b198ca52e6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

47 FITS frames (31 MB), WASP-140190123020043.FITS → WASP-140190123042112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-190123.log (74a405408145...), FinalLightCurve_WASP-140 b_2019-01-22.png (7a6df098a722...), FinalLightCurve_WASP-140 b_2019-01-22.csv (9d469b16703d...)

Compute resources

Wall time 185.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 01:29Z.